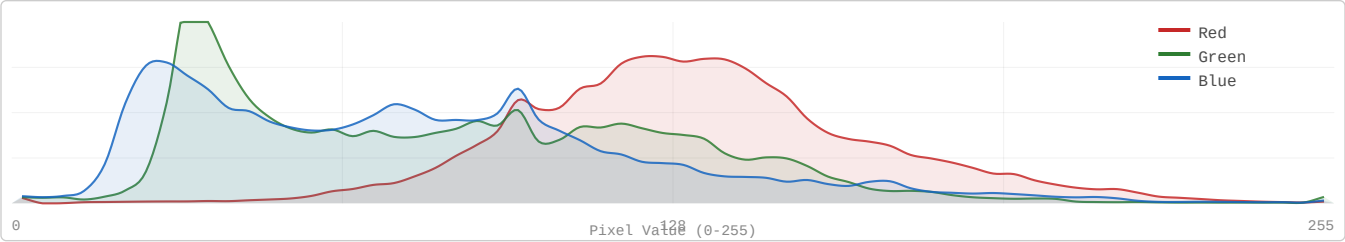
Decorrelation Gap: KODIM04

512x768 | 24-bit RGB | PNG (lossless)

Red-Roofed Building by Water | Weakly one-dimensional (PC1 = 86.87%)

1. RGB Channel Distribution



2. Inter-Channel Correlation: RGB vs YCbCr

RGB (Original)

	R	G	B
R	1.0000	0.6004	0.6850
G	0.6004	1.0000	0.9568
B	0.6850	0.9568	1.0000

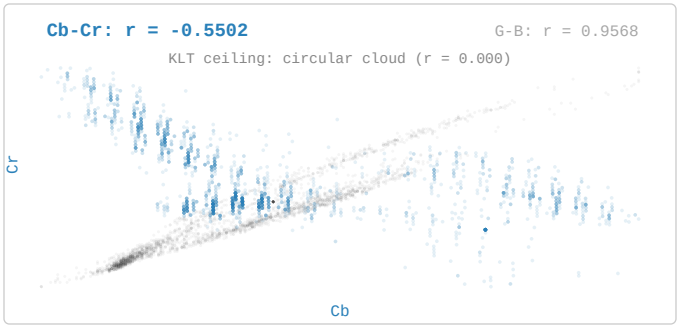
Cb-Cr: -0.550

(KLT ceiling = 0.000)

YCbCr (BT.601)

	Y	Cb	Cr
Y	1.0000	0.4880	-0.4609
Cb	0.4880	1.0000	-0.5502
Cr	-0.4609	-0.5502	1.0000

3. Cb vs Cr Scatter with G-B background overlay



4. Pairwise Decorrelation Gap

Channel Pair	RGB	YCbCr (BT.601)	Residual |r|
R-G → Y-Cb	0.600446	0.488011	0.4880
R-B → Y-Cr	0.684965	-0.460854	0.4609
G-B → Cb-Cr	0.956832	-0.550224	0.5502
Avg |r|	0.747414	0.499696	0.4997

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5. Decorrelation Efficiency & Blue Independence

Decorrelation Efficiency

33.1% Low

0% BT.601 leaves all — 100% KLT ceiling

Blue Channel Independence (Baetzel 2026a)

2.7%

0% Fully predictable — 100% Fully independent

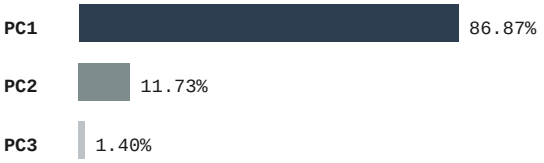
6. 4:2:0 Chroma Subsampling — Information Cost

Channel	PSNR (dB)
R	40.97 dB
G	46.78 dB
B	46.17 dB
Overall	43.80 dB

Method: BT.601 RGB→YCbCr | 2x box downsample Cb,Cr | nearest upsample | BT.601 inverse

7. PCA Baseline Reference (Baetzel 2026a)

Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.3905	-0.6332	-0.6682	4473.64
PC2	-0.9085	0.3824	0.1685	604.25
PC3	-0.1488	-0.6729	0.7246	71.90

8. PCA Derived Metrics

Condition Number

62.22

PC1/PC2: 7.40 | PC2/PC3: 8.40

Blue Channel Independence

2.7%

Variance not captured by PC1

PC1 Dominant Channel

B

Loading: B (0.67)

9. Redundancy Profile

Classification: Weakly one-dimensional | CN: 62.22 | PC1: 86.9%

RGB Correlation: Avg |r| = 0.747414 | R-G: 0.6004 | R-B: 0.6850 | G-B: 0.9568

YCbCr Residual: Avg |r| = 0.499696 | Y-Cb: 0.4880 | Y-Cr: -0.4609 | Cb-Cr: -0.5502

Efficiency: 33.1% of KLT ceiling | Gap: 0.4997 avg |r| remaining

Blue Channel: 2.7% independent | 4:2:0 PSNR: 43.80 dB

References

[1] Baetzel, J. (2026a). "Statistical Characterization of Inter-Channel Redundancy Structure"
[2] Wallace, G.K. "The JPEG still picture compression standard." IEEE Trans. CE 38(1), 1992.
[3] ITU-R Recommendation BT.601: Studio Encoding Parameters, 1982.
[4] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.